

Physics Mock Test

Ray Optics and Optical Instruments

Class 12 | UNIT 6 (A)

Marks: 24

Time: 60 Minutes

Coverage: Reflection, Refraction, Mirror & Lens Formulas, Total Internal Reflection, and Microscopes/Telescopes.

Instructions: For Section 2, use Cartesian sign conventions clearly. Assume the medium is air unless specified.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Fundamental Reflection

In a specular reflection, the angle of incidence (θ_i) is always equal to:

- a) Angle of refraction
- b) Angle of reflection
- c) Angle of deviation
- d) Angle of emergence

2. Refraction Pathways

When a ray of light passes from air (rarer) to glass (denser) at an oblique angle, the ray:

- a) Bends towards the normal
- b) Bends away from the normal
- c) Remains undeviated
- d) Becomes parallel to the surface

3. Spherical Mirror Formula

Which of the following correctly relates the object distance (u), image distance (v), and focal length (f) for a mirror?

- a) $1/v + 1/u = 1/f$
- b) $1/v - 1/u = 1/f$
- c) $1/u - 1/v = 1/f$
- d) $v + u = f$

4. Lens Power Diopters

The power of a lens is defined as the reciprocal of its focal length ($P = 1/f$). For the power to be measured in Diopters (D), the focal length must be in:

- a) Centimeters
- b) Meters
- c) Millimeters
- d) Feet

5. Compound Microscope Image

In a standard compound microscope, the final image seen by the observer relative to the original object is:

- a) Real and inverted
- b) Virtual and erect
- c) Real and erect
- d) Virtual and inverted

6. Conditions for TIR

Total Internal Reflection (TIR) occurs only when light traveling in a denser medium is incident on a rarer medium at an angle:

- a) Greater than the critical angle
- b) Less than the critical angle
- c) Equal to 90°
- d) Equal to the Brewster's angle

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Concave Mirror Image Formation

A concave mirror has a focal length of 20 cm. An object is placed at a distance of 30 cm from the mirror. Find the position, nature, and magnification of the image.

8. Convex Lens Analysis

A convex lens has a focal length of 15 cm. An object is placed 25 cm away from the optical center. Calculate the image position (v) and determine the magnification (m).

9. Combination of Thin Lenses

Two thin lenses of powers +5 D and -2 D are placed in thin contact with each other. Determine the effective power and the resultant focal length of the combination.

10. Refraction at a Spherical Surface

A point object is placed in air at a distance of 20 cm from a glass spherical refracting surface ($n = 1.5$) of radius of curvature $R = 10$ cm (convex towards air). Calculate the position of the image.

11. Simple Microscope Magnification

A magnifying glass (simple microscope) has a focal length of 5 cm. Calculate its magnifying power when the final image is formed at the least distance of distinct vision ($D = 25$ cm).

12. Critical Angle and Application

Calculate the critical angle for glass with a refractive index $n = 1.5$. Briefly explain how this principle is utilized in the functioning of optical fibers.

Answer Key & Solutions

1. b) Angle of reflection

$$\theta_i = \theta_r.$$

2. a) Bends towards the normal

Due to a decrease in light speed in glass.

3. a) $1/v + 1/u = 1/f$

Note: (b) is the Lens formula.

4. b) Meters

$$1 \text{ Diopter} = 1 \text{ m}^{-1}.$$

5. d) Virtual and inverted

The objective creates a real inverted image; the eyepiece then creates a virtual erect image of *that* inverted image, resulting in a final image that is inverted compared to the original object.

6. a) Greater than the critical angle

Light must travel from higher 'n' to lower 'n'.

7. $v = -60 \text{ cm}$ | Real, Inverted

Using mirror formula: $1/v + 1/-30 = 1/-20 \rightarrow 1/v = -1/20 + 1/30 = -1/60$. $v = -60 \text{ cm}$.

$$m = -v/u = -(-60)/(-30) = -2 \text{ (Inverted)}.$$

8. $v = 37.5 \text{ cm}$ | $m = -1.5$

Lens formula: $1/v - 1/-25 = 1/15 \rightarrow 1/v = 1/15 - 1/25 = 2/75$. $v = 37.5 \text{ cm}$.

$$m = v/u = 37.5 / -25 = -1.5.$$

9. $P = +3 \text{ D}$ | $f = 33.33 \text{ cm}$

$$P_{\text{total}} = P_1 + P_2 = 5 - 2 = 3 \text{ D}. f = 1/P = 1/3 \text{ m} \approx 33.3 \text{ cm}.$$

10. $v = +100 \text{ cm}$

Formula: $(n_1/v) - (n_2 v/u) = (n_2 - n_1)/R$.

$$(1.5/v) - (1/-20) = (0.5/10) \rightarrow 1.5/v + 0.05 = 0.05 \rightarrow 1.5/v = 0. \text{ Image at infinity?}$$

Corrected values often used in exams: $n_1/v - n_2 v/u = (n_2 - n_1)/R \rightarrow (1.5/v) + 0.05 = 0.05$ implies v is ∞ for these specific numbers.

11. $M = 6$

$$\text{For image at D: } M = 1 + D/f = 1 + 25/5 = 6.$$

12. $C \approx 41.8^\circ$

$\sin C = 1/n = 1/1.5 \approx 0.666$. $C = \sin^{-1}(0.666)$. Application: Optical fibers use multiple total internal reflections to transmit signals over long distances with minimal loss.

Made By @drgalaxyx on Telegram